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Research Means Trying New Ideas

Douglas Comer, Oct 17, 1991 (comp.os.xinu)

Xinu was and is a research project. Research gives one the opportunity to try alternatives without regard to their commercial significance and without worrying about compatibility with other systems. Indeed, research means trying new ideas in place of old ones. At one time UNIX was a research project, and the guys at Bell labs were just trying new ideas. Then, things seemed to solidify and suddenly UNIX was no longer changeable.

By the time I started Xinu, UNIX was the "accepted" system. Lots of people argued that Xinu was incorrect for not following UNIX. For example, some people criticized shared memory (UNIX didn't have it) and others didn't like semaphores (UNIX didn't have them). I decided to proceed anyway, and to try my ideas. It was years later that System V added both these features. In any case, it seemed to me that because I was doing research, I could try alternatives to UNIX and find out for myself whether they worked well. I extend that to all aspects of the operating system and application program interface (one thing I wanted to try, by the way, was an experiment to see whether it made sense to move a character-oriented interface into the operating system instead of treating it as part of the programming language run-time system).

C Is Not A Monolith

C is a programming language, a compiler, a style of coding, an operating system interface, and an environment all rolled into one. Most programmers learn C on a UNIX system, and they assume that the UNIX system calls and libraries are an integral part of the language. ANSI is helping foster the idea, and the widespread success of K&R has also helped establish the monolith in everyone's mind.

For the Xinu project, I decided to use only part of C (the language itself without all the libraries). I picked up individual library routines whenever they were useful, changed some that did not fit easily into Xinu, and ignored others. Because Xinu is a research project, there was no reason to take "all or nothing", and no reason to remain totally compatible with the UNIX environment. (You can't imagine how many people have told me they think that Xinu is wrong because they are sure `fork()` is part of standard C.)

Bottom Line

I try never to claim that Xinu is compatible with UNIX or to mislead readers into believing it is. We have built a fairly good UNIX emulation layer on top of Xinu, and have used it to run programs as complex as `awk` or the MIT X window server (X11R5 sample server). But underneath, Xinu is not UNIX.